

E. G. S. PILLAY ENGINEERING COLLEGE
(Autonomous)



Approved by AICTE, New Delhi| Affiliated to Anna University, Chennai
Accredited by NAAC with A++Grade| Accredited by NBA T1
NAGAPATTINAM – 611 002

B.E. Biomedical Engineering

Full Time Curriculum and Syllabus

Third Year – Sixth Semester

COURSE CODE	COURSE NAME	CATEGORY	L	T	P	C	MAX.MARKS		
							CA	ES	TOTAL
THEORY COURSES									
2302BM601	Digital Image Processing	PCC	3	0	0	3	40	60	100
2302BM602	Bio Embedded Systems	PCC	3	0	0	3	40	60	100
2302BM603	Biomedical IoMT	PCC	3	0	0	3	40	60	100
	Professional Elective III	PEC	3	0	0	3	40	60	100
	Professional Elective IV	PEC	3	0	0	3	40	60	100
2303BM038	Open Elective II	OEC	3	0	0	3	40	60	100
2304MC605	Mandatory course-II	MC	3	0	0	0	100	0	100
PRACTICAL COURSES									
2302BM651	Image Processing Laboratory	PCC	0	0	2	1	60	40	100
2302BM652	Bio Embedded Laboratory	PCC	0	0	2	1	60	40	100
2302BM653	Mini Project	EEC	0	0	2	1	50	50	100
2304GE601	Professional Development Course-IV	EEC	0	0	2	1	100	--	100
TOTAL			18	00	8	22	610	490	1100

L–Lecture| T–Tutorial| P–Practical| C–Credit| CA–Continuous Assessment| ES–End Semester

2302BM601	DIGITAL IMAGE PROCESSING								L	T	P	C	
									3	0	0	3	
PREREQUISITE:													
Bio signal processing													
COURSE OBJECTIVES:													
	To introduce the basic principles of image acquisition, color image processing, and image transform techniques used in digital image processing systems.												
	To develop the ability to pre-process digital images using enhancement and filtering techniques for improving image quality and interpretability.												
	To provide knowledge of region-based image segmentation methods, with emphasis on region growing techniques for extracting homogeneous image regions.												
	To familiarize learners with shape feature extraction methods for object identification and image classification applications.												
	To impart understanding of image processing techniques in various medical application.												
COURSE OUTCOMES: After completion of the course students able to													
CO1	Understand basic principles of image acquisition, process color images and apply image transforms												
CO2	Pre-process the images using image enhancement and filtering techniques												
CO3	Apply region-based segmentation techniques such as region growing to extract homogeneous regions from images												
CO4	Determine shape features of objects in images to support object identification and classification.												
CO5	Give Examples of medical applications and ongoing developments												
COs Vs POs MAPPING:													
	COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	
	CO1	3	3	3	2	2			1		1	1	
	CO2	3	3	3	2	2			1		1	1	
	CO3	3	3	3	2	2			1		1	1	
	CO4	3	3	3	2	2			1		1	1	
	CO5	3	3	3	2	2			1		1	1	
COs Vs PSOs MAPPING													
COURSE CONTENTS:													
Module I		DIGITAL IMAGE FUNDAMENTALS								9 Hours			
Elements of digital image processing systems, Image sensing and Acquisition, - Elements of visual perception, Mach band effect, Color image fundamentals - RGB, HSI models, Image sampling, Quantization, dither, Two-dimensional mathematical preliminaries, 2D transforms - DFT, DCT, KLT, SVD.													
Module II		IMAGE ENHANCEMENT AND RESTORATION								9 Hours			
Point processing, Histograms, Histogram equalization and specification techniques, Noise distributions, Spatial averaging, Directional Smoothing, Median, Geometric mean, Harmonic mean, Contra harmonic mean filters, Homomorphic filtering, Color image enhancement. Image Restoration- degradation model, Unconstrained and Constrained restoration. Geometric													

transformations.		
Module III	IMAGE SEGMENTATION	9 Hours
Edge detection, Edge linking via Hough transform – Thresholding - Region based segmentation – Region growing – Region splitting and merging – Segmentation by morphological watersheds – Hybrid methods – Image Registration		
Module IV	IMAGE FEATURE EXTRACTION AND SELECTION	9 Hours
Features- Feature space, Statistical features, Texture features-co-occurrence features and Run length features, shape features. Feature selection - Need - PCA, statistical analysis and selection of features.		
Module V	MEDICAL APPLICATIONS AND ON GOING DEVELOPMENTS	9 Hours
Medical applications of imaging: Computer-aided diagnosis in mammography-Tumor imaging and treatment -Angiography Bone strength and osteoporosis - Frontiers of image processing in medicine-Trends		
		Total: 45 Hours
Further Readings:	Applications in medical image analysis -Computer-based activities	
Text Books:		
1. Rafael C. Gonzalez, Richard E. Woods, “Digital Image Processing”, Pearson, Education, Inc., 4th Edition, 2017.		
2. Anil K. Jain, “Fundamentals of Digital Image Processing”, Pearson Education, Inc., 1st Edition, 2015.		
3. Milan Sonka, Vaclav Hlavac and Roger Boyle, “Image Processing, Analysis and Machine vision”, Cengage, 3rd Edition, 2013.		
References:		
1. Kenneth R. Castleman, "Digital Image Processing", Pearson, 2006.		
2. Rafael C. Gonzalez, Richard E. Woods, Steven Eddins, "Digital Image Processing using MATLAB", Pearson Education, Inc., 2004.		
3. William K. Pratt, "Digital Image Processing", John Wiley, New York, 2002.		

2302BM602	BIO EMBEDDED SYSTEMS							L	T	P	C	
								3	0	0	3	
PREREQUISITE:												
Analog and digital devices												
COURSE OBJECTIVES:												
	1. To introduce the architecture, instruction set, and programming fundamentals of the 8086 microprocessor and microcontrollers.											
	2. To provide foundational knowledge on embedded system concepts, categories, recent trends, and development workflows.											
	3. To familiarize students with key embedded communication protocols such as UART, I2C, SPI, ADC, PWM, and Timers.											
	4. To understands the RTOS and inter-process communication.											
	5. To develop the ability to apply embedded system concepts in designing medical and healthcare applications including monitoring, wearable, and implantable devices.											
COURSE OUTCOMES:												
CO1	Explain the architecture, instruction set and programming concepts of 8086 microprocessor.											
CO2	Illustrate the structure, specialties, and development workflow of embedded systems, including hardware/software architecture and communication software.											
CO3	Analyze embedded communication protocols (UART, SPI, I2C, ADC, PWM, Timers) for real-time biomedical applications.											
CO4	Examine the RTOS and inter-process communication.											
CO5	Evaluate embedded solutions for healthcare systems including monitoring devices, wearables, and implantable systems while considering safety and performance constraints.											
COs Vs POs MAPPING:												
	COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
	CO1	2	1	2	-	2	2	-	-	-	-	2
	CO2	2	2	1	2	2	-	-	-	-	-	2
	CO3	2	2	2	3	2	-	-	-	-	-	2
	CO4	2	2	2	1	2	-	2	-	-	-	2
	CO5	2	2	2	-	2	-	-	-	-	-	2
	Avg	2	1.8	1.8	2	2	2	2				2
COs Vs PSOs MAPPING												

Embedded System-Definition, Categories of embedded systems, Specialties of embedded systems, Recent trends in embedded systems, Block diagram of embedded system, Hardware architecture, Software architecture, Communication software, Quality attributes of an Embedded systems		
MODULE III	PERIPHERAL INTERFACING	9 Hours
I/O Devices, UART, ADC interfacing for biosignals (ECG/PPG/Temperature), PWM, Timer, Interfacing Protocols-I2C, SPI		
MODULE IV	RTOS BASED EMBEDDED SYSTEM DESIGN	9 Hours
Operating system basics, types of operating systems, tasks, process and threads, multiprocessing and multitasking, task scheduling: non-pre-emptive and pre-emptive scheduling; task communication-shared memory, message passing.		
MODULE V	EMBEDDED SYSTEMS IN HEALTHCARE	9 Hours
Design and development of embedded systems for medical devices, Application of embedded systems in patient monitoring, Wearable health monitoring systems, Implantable medical devices, and Challenges and considerations for medical embedded systems.		
Total:		45Hours
Further readings		
Text Books:		
1.Ray K& Bhurchandi K.M," Advanced Microprocessors and Peripherals: Architecture, Programming and Interface", 3rd Edition, McGraw Hill, New Delhi, 2012.		
2. Prasad KVK K Embedded Real Time Systems Concepts, Design and Programming Dream tech press 2013		
3. Muhammad Ali Mazidi ,Sarmad Naimi and Sepehr Naimi TIARM Peripherals Programming and Interfacing Using C Language Pearson Education 2014		
References:		
1.JonathanW.ValvanoEmbeddedSystems Introduction to Arm CortexM Microcontrollers 5th Edition ISBN 978-1477508992 2014		
2. 2. Embedded System Design: Embedded Systems Foundations of Cyber-Physical Systems, and the Internet of Things , 4th Edition , ISBN 978-3030609108, 2021		

2302BM603	BIOMEDICAL IoMT						L	T	P	C		
	(B.E-BME)						3	0	0	3		
PREREQUISITE:												
	Biomedical Instrumentation, Digital Electronics, Basic Communication Systems											
COURSE OBJECTIVES:												
	1. To understand the fundamentals and architecture of IoT and its relevance to biomedical applications.											
	2. To learn embedded systems, sensors, and actuators used in IoT-based healthcare devices.											
	3. To study communication technologies and protocols for biomedical IoT systems.											
	4. To understand cloud, edge computing, and data management for healthcare IoT applications.											
	5. To analyze healthcare IoT use cases, challenges, security, and regulatory aspects											
COURSE OUTCOMES:												
On the successful completion of the course the students will be able to,												
CO1	Explain IoT concepts, architecture, and components used in biomedical applications.											
CO2	Design basic embedded IoT systems using biomedical sensors and actuators.											
CO3	Implement suitable communication protocols for biomedical data transmission.											
CO4	Utilize cloud and edge computing platforms for healthcare data storage and analytics.											
CO5	Apply IoT principles to solve real-world healthcare problems considering security and regulatory issues.											
COs Vs POs MAPPING:												
COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	
CO1	3	2	1	1	2	–	–	–	–	–	1	
CO2	3	2	2	1	2	–	–	–	–	–	1	
CO3	3	2	2	1	2	–	–	–	–	–	1	
CO4	3	2	1	1	2	–	–	–	–	–	1	
CO5	3	2	1	1	2	–	–	–	–	–	1	
COs Vs PSOs MAPPING												
				COs	PSO1	PSO2	PSO3					
				CO1	2	–	1					
				CO2	2	–	1					
				CO3	2	–	1					
				CO4	2	–	1					
				CO5	2	–	1					
COURSE CONTENTS:												
Module I	FUNDAMENTALS OF IoT									9 Hours		
What is IoT: definitions and evolution. Enabling technologies: sensors, actuators, and embedded systems. Physical and logical design of IoT. IoT architecture frameworks and layered models. IoT system components and ecosystem. Introduction to Internet of Medical Things (IoMT) and healthcare applications.												
Module II	EMBEDDED SYSTEMS FOR IoT									9 Hours		

Microcontrollers and microprocessors. Embedded platforms: Arduino and Raspberry Pi. Hardware and software architecture. Sensor interfacing and data acquisition. Actuators and control systems. Basics of real-time operating systems. Energy management in embedded IoT devices.		
Module III	COMMUNICATION TECHNOLOGIES AND PROTOCOLS	9 Hours
Networking fundamentals: LAN, WAN, and PAN. Wireless protocols: Wi-Fi, Bluetooth, Zigbee, and LoRaWAN. IoT protocol stack. MQTT, CoAP, and HTTP for IoT applications. IPv6 and 6LoWPAN for low-power networks		
Module IV	DATA MANAGEMENT AND CLOUD INTEGRATION	9 Hours
Data formats, APIs, and messaging. Data storage and retrieval. Cloud platforms for healthcare IoT. Fog and edge computing concepts. Security, privacy, and ethical considerations of biomedical IoT data.		
Module V	IoT IN HEALTHCARE	9 Hours
Wearable health monitoring systems. Remote patient monitoring. Telemedicine support systems. Smart medical devices. Case studies on sensor networks for clinical applications. Challenges and regulatory aspects in healthcare IoT.		
TOTAL:		45 Hours
FURTHER READING:		
Text books:		
1. Arshdeep Bahga, Vijay Madisetti, <i>Internet of Things: A Hands-on Approach</i> , Universities Press.		
2. Rajkumar Buyya, Amir Vahid Dastjerdi, <i>Internet of Things: Principles and Paradigms</i> , Morgan Kaufmann		
3. Valentina Emilia Balas, Souvik Pal, <i>Healthcare Paradigms in the Internet of Things Ecosystem</i> , Academic Press		
References:		
1. R. S. Khandpur, <i>Handbook of Biomedical Instrumentation</i> , McGraw-Hill.		
2. Fagen Li et al., <i>IoT Technologies for Healthcare</i> , Springer.		
3. U. Mitra, <i>Wireless Body Area Networks</i> , IEEE Press		

2302BM651	IMAGE PROCESSING LABORATORY	L	T	P	C
	(B.E-BME)	0	0	2	1

PREREQUISITE:

Bio signal processing

COURSE OBJECTIVES:

To introduce fundamental concepts and techniques of digital image processing, including image acquisition, enhancement, filtering, and segmentation.

To develop analytical and computational skills required to process images in spatial and frequency domains for effective interpretation.

To provide practical understanding of feature extraction, morphological operations, and region-based methods for meaningful image analysis and decision-making.

COURSE OUTCOMES:

On the successful completion of the course the students will be able to,

CO1	Apply image sampling, quantization, enhancement, and segmentation techniques to process digital images.
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C02	Analyze spatial and intensity resolution, histogram characteristics, and frequency components of digital images.
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CO3	Compute intensity transformations, threshold values, and frequency-domain representations for image analysis.
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C04	Determine appropriate spatial, frequency domain filtering, and segmentation methods for given image processing problems
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C05	Apply morphological operations and region-based segmentation techniques to extract meaningful image structures.
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COs Vs POs MAPPING:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	2	1	2	2	—	—	—	—	1	2
CO2	2	3	1	2	2	—	—	—	—	1	2
CO3	3	2	1	2	3	—	—	—	—	1	2
CO4	2	3	2	2	2	—	—	—	—	1	2
CO5	2	2	2	2	3	—	—	—	-	1	2

COs Vs PSOs MAPPING

COs	PSO1	PSO2	PSO3
CO1	2	1	1
CO2	2	2	1
CO3	2	3	1
CO4	3	2	2
CO5	3	2	3

COURSE CONTENTS:

LIST OF EXPERIMENTS

1. Perform and analyze image sampling and quantization processes.
2. Analyze and evaluate the effects of spatial and intensity resolution on image
3. Implement and analyze intensity transformation techniques
4. Perform and analyze Discrete Fourier Transform (DFT) operations for frequency-domain image analysis.
5. Execute and analyze histogram processing methods and basic thresholding techniques
6. Implement and analyze spatial domain filtering techniques for image enhancement.
7. Perform and analyze frequency-domain filtering methods to enhance and restore images.

8. Implement and analyze edge detection techniques for identifying image boundaries.	
9. Perform and analyze line detection methods for extracting linear structures from images.	
10. Execute and analyze point detection techniques to identify isolated features in digital images.	
11. Use and analyze morphological operations for image structure analysis and shape manipulation.	
12. Implement and analyze region-based segmentation techniques	
TOTAL:	30 Hours
Further Reading: 1. Color image segmentation algorithm development	
References:	
1. William K. Pratt, "Digital Image Processing", John Wiley, New York, 2002.	
2. Alan C. Bovik, "Handbook of image and Video Processing", Elsevier Academic press, 2005.	
3. MATLAB / Python with Open CV documentation and medical image datasets (NIH	

2302BM653	MINI PROJECT						L	T	P	C	
	(B.E-BME)						0	0	2	1	
PREREQUISITE:											
	Knowledge of all subjects related to biomedical										
COURSE OBJECTIVES:											
	1.Identify the problem statement and apply the engineering concepts to find the solution.										
	2.Improve the analyzing capability of the students.										
	3.Increase the exuberance in finding the solution to various problems.										
COURSE OUTCOMES:											
On the successful completion of the course the students will be able to,											
CO1	Formulate a real-world problem, identify the requirement, and develop the design solutions.										
CO2	Identify technical ideas, strategies, and methodologies.										
CO3	Utilize the new tools, algorithms, and techniques that contribute to obtaining the solution of the project.										
CO4	Test and validate through conformance of the developed prototype and analysis of the cost- effectiveness.										
CO5	Prepare the report and present oral demonstrations										
COs Vs POs MAPPING:											
COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	1	2	1	1	2				2	2	2
CO2	1	2	1	1	2				2	2	2
CO3	1	2	1	1	2				2	2	2
CO4	1	2	1	1	2				2	2	2
CO5	1	2			2				2	2	2
COs Vs PSOs MAPPING											
COs	PSO1	PSO2	PSO3								
CO1	1	1	1								
CO2	1	1	1								
CO3	1	1	1								
CO4	1	1	1								
CO5	1	1	1								

2302BM652	BIO-EMBEDDED LABORATORY	L	T	P	C
		0	0	2	1

PREREQUISITE:

	Devices and Circuits Lab
	Applied Digital Logic and Design

COURSE OBJECTIVES:

	To perform the arithmetic and logical operations in 8051.
	To introduce interfacing, I/o Devices to the Processor.
	To introduce the evolution of the Internet of Things (IoT).
	To apply the concept of the Internet of Things in real-world scenario.

COURSE OUTCOMES:

On the successful completion of the course, students will be able to	
CO1:	Understand the arithmetic and logical operation in 8051 microcontroller.
CO2:	Learn the ARM processor with Interfacing of ADC and DAC.
CO3:	Learn the ARM processor with Interfacing of LED and LCD.
CO4:	Design an Embedded based system with real time application.
CO5:	Design an IoT based system with real time application.

COs Vs POs MAPPING:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	3	3	2	2					1	1
CO2	3	3	2	2	2					1	1
CO3	3	3	2	2	2					1	1
CO4	3	3	2	2	2					1	2
CO5	3	3	3	3	3					1	2

COs Vs PSOs MAPPING:

COs	PSO 1	PSO 2	PSO3
CO1	3	2	
CO2	3	3	
CO3	3	3	
CO4	3	3	
CO5	3	3	

LIST OF EXPERIMENTS:

Experiments using 8051.

1. Implementation of Arithmetic and Logical Operations using 8051 for Signal Processing	3 Hours
2. Generation of Square Waveforms using 8051 for Timing and Control Applications	3 Hours

Experiments Using ARM

3. Interfacing of Analog-to-Digital Converter (ADC) and Digital-to-Analog Converter (DAC) using ARM for Signal Acquisition and Reconstruction	3 Hours
	3 Hours

4. Control of LED Indicators using ARM Microcontroller for Equipment Status Display	
5. Interfacing of LCD Module using ARM Microcontroller for Display of various Parameters	3 Hours
6. Interfacing of Keyboard / Keypad using ARM Microcontroller for various Data Input	3 Hours
7. Interfacing and Control of Stepper Motor using ARM Mini Projects for Embedded and IOT	3 Hours
8. Design and Development of an Automatic Night-Assist Lighting System using LDR	3 Hours
9. Smart Greenhouse-Plant Monitoring System	3 Hours
10. Object Detection and Alert System for Safety Applications	3 Hours
TOTAL: 30 HOURS	
Experiments are based on the Industry curriculum. Syllabus drawn from the Hitakey Technology (Embedded Systems Experiments) and Techknots (IoT Experiments).	
REFERENCES:	
1. B.Arunkumar, “Embedded and Iot design Laboratory Manual”, 2021.	
2. https://www.studocu.com/in/document/mount-litera-zee-school/computer-science-sl/cs3691-embedded-systems-and-iot-lab-manual/88752899 .	
3. https://www.studocu.com/in/document/vel-tech-rangarajan-dr-sagunthala-rd-institute-of-science-and-technology/embedded-system-design/es-iot-lab-manual/84437016	
4. https://www.slideshare.net/slideshow/embedded-systems-and-iot-lab-manual-for-engineering-students/267160236	

PROFESSIONAL ELECTIVE - III

SL. NO.	COURSE CODE	COURSE TITLE	CATEGORY	PERIODS PER WEEK			TOTAL CONTACT PERIODS	CREDITS
				L	T	P		
1.	2303BM014	Radiological Equipment's	PEC	3	0	0	3	3
2.	2303BM015	Patient safety, Standards and Ethics	PEC	3	0	0	3	3
3.	2303BM016	Medical Waste Management	PEC	3	0	0	3	3
4.	2303BM017	Physiological Modeling	PEC	3	0	0	3	3
5.	2303BM018	Pattern recognition techniques	PEC	3	0	0	3	3
6.	2303BM019	Body Area Networks	PEC	3	0	0	3	3

2303BM014		RADIOLOGICAL EQUIPMENTS	L	T	P	C
			3	0	0	3

PREREQUISITE:

Human Anatomy & Physiology, and Biomedical Instrumentation

Familiarity with Imaging Principles

COURSE OBJECTIVES:

- 1.To get the clear understanding of X-ray generation, radio isotopes and various techniques used for Visualizing organs..
- 2.To study about the functioning of X-ray tubes and method by which fogginess scan be reduced
- 3.To study about the different types radio diagnostic unit, transverse tomography and types of radio detection.
- 4.To understand the function of X ray generation and radioisotopes.
- 5.To know the concepts of MRI functionality and imaging various sections of body.

COURSE OUTCOMES:

Upon successful completion of this course, students will be able to:

- | | |
|------------|--|
| CO1 | Explain the different radio diagnostic and therapeutic techniques |
| CO2 | Illustrate the principle computed tomography. |
| CO3 | Interpret the technique used for visualizing various sections of the body using magnetic resonance imaging |
| CO4 | Demonstrate the applications of radion uclide imaging. |

COs Vs POs MAPPING:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	2	1	-	-	-	-	-	-	-	1
CO2	3	2	1	-	-	-	-	-	-	-	1
CO3	3	2	1	-	-	-	-	-	-	-	1
CO4	3	2	1	-	-	-	-	-	-	-	1
CO5	3	2	1	-	-	-	-	-	-	-	1

COs Vs PSOs MAPPING:

COs	PSO1	PSO2	PSO3
CO1	2	2	-
CO2	2	2	-
CO3	2	2	-
CO4	2	-	-
CO5	-	-	-
AVG	2	2	-

Module I	X–RAY MACHINES & DIGITAL RADIOGRAPHY	9 Hours
Basis of diagnostic radiology, Nature of X-Rays, Production of X-Rays, X-Ray machine, visualization of X-Rays, dental X-Ray machines, portable and mobile X-ray units, physical parameters for X-Ray detectors, digital radiography.		
Module II	COMPUTER TOMOGRAPHY	9Hours
Principles of tomography, CT Generations, X- Ray sources- collimation- X- Ray detectors-Viewing systems- spiral CT scanning – Ultra fast CT scanners. Advantages of computed radiography over film screen radiography: Time, Image quality, Lower patient dose, Differences between conventional		

imaging equipment and digital imaging equipment: Image plate, Plate readers, Image characteristics, Image reconstruction techniques- back projection and iterative method. Spiral CT, 3D Imaging and its application.		
Module III	MAGNETIC RESONANCE IMAGING & SPECTROCOPY	9Hours
Fundamentals of magnetic resonance- Interaction of Nuclei with static magnetic field and Radio frequency wave- rotation and precession – Induction of magnetic resonance signals – bulk magnetization – Relaxation processes T1 and T2. Block Diagram approach of MRI system magnet (Permanent, Electromagnet and Super conductors),generations of gradient magnetic fields, Radio Frequency coils (sending and receiving),and shimcoils, Electronic components, MRI.		
Module IV	NUCLEAR MEDICAL IMAGING SYSTEMS	9Hours
Nuclear imaging – Anger scintillation camera –Nuclear tomography – single photon emission computer tomography, positron emission tomography – Recent advances .Radionuclide imaging, Bone imaging, dynamic renal function ,myocardial per fusion. Non imaging technique ,shematological measurements, Glomerular Filtration rate, volume measurements, clearance measurement, whole-body counting, surface counting		
Module V	HUMAN RADIOBIOLOGY	9Hours
Radiation therapy – linear accelerator, Telegamma Machine. SRS –SRT,-Recent Techniques in radiation therapy - 3DCRT – IMRT – IGRT and Cyber knife- radiation measuring instruments, Dosimeter, film badges, Thermo Luminescent dosimeters- electronic dosimeter-Radiation protection in medicine-radiation protection principles.		
		Total: 45Hours
FURTHER READING: Radiation Therapy		
Text book:		
1.Bushberg J. T., Seibert J. A., Leidholdt Jr. E. M., Boone J. M., <i>The Essential Physics of Medical Imaging</i> , 3rd Edition, Lippincott Williams & Wilkins, 2012		
2.Haidekker, Mark A., <i>Medical Imaging Technology</i> , Springer, 2013		
3.Seeram, Euclid, <i>Computed Tomography: Physical Principles, Clinical Applications, and Quality Control</i> , 4th Edition, Saunders, 2015		
Reference Books:		
1.Carlton R. R. & Adler A. M., <i>Principles of Radiographic Imaging: An Art and a Science</i> , 6th Edition, Cengage Learning, 2019.		
2.Farr R. F. & Allisy A., <i>Physics for Medical Imaging</i> , Saunders, 1997.		
3.Khandpur R. S., <i>Handbook of Biomedical Instrumentation</i> , 3rd Edition, McGraw-Hill Education, 2014.		
4.Cherry, Sorenson & Phelps, <i>Physics in Nuclear Medicine</i> , 4th Edition, Elsevier, 2012.		
5.Hendee W. R. & Ritenour E. R., <i>Medical Imaging Physics</i> , 4th Edition, Wiley-Liss, 2002		

2303BM015	PATIENT SAFETY, STANDARDS AND ETHICS (For B.E-BME)					L	T	P	C		
						3	0	0	3		
PREREQUISITE:											
	Basic Medical Sciences										
	Healthcare Systems & Practice, Ethical Foundations										
COURSE OBJECTIVES:											
	1.To introduce the safety procedures in healthcare organizations										
	2.To familiarize the healthcare organization structure and responsibilities										
	3.To explore the safety standard to be followed in hospitals										
	4.To understand the role of ethics in medicine										
	5.To Recognize ethical issues when they arise in your practice										
COURSE OUTCOMES:											
On the successful completion of the course the students will be able to,											
CO1	Apply safety procedures in healthcare organizations										
CO2	Analyze the Health care organization structure and the responsibilities of different levels to implement safety										
CO3	Analyze the regulatory standards for medical device maintenance										
CO4	Outline the accreditation protocols for a hospital and its safety standards										
CO5	Identify ethical issues in medicine, health care and life sciences										
COs Vs POs MAPPING:											
COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	2	2	-	1	-	3	2	2	-	-	1
CO2	2	2	-	2	-	2	2	2	-	-	1
CO3	2	3	2	2	-	3	2	2	-	-	1
CO4	1	2	1	2	-	2	2	2	-	-	1
CO5											1
COs Vs PSOs MAPPING											
				COs	PSO1	PSO2	PSO3				
				CO1	2	2	-				
				CO2	2	2	-				
				CO3	3	2	2				
				CO4	2	2	2				
				CO5	2	2	-				
COURSE CONTENTS:											
Module I		EFFECTS OF ELECTRICITY							9 Hours		
Physiological effects of electricity – important susceptibility parameters – micro shock – macro shock hazards -patients electrical environment – isolated power system – conductive surfaces											
Module II		PATIENT SAFETY LAWS AND REGULATIONS							9 Hours		
Mandatory Reporting systems - Anatomy of a patient safety Law: Compliance Tips - Federal patient safety											

Legislation Initiatives- Medical Device Reporting- Clinical trials and Adverse-Event Reporting - Patient safety Goals and standards- The Quality Assessment and performance Improvement rule		
Module III	PATIENT SAFETY IN MAIN CLINICAL SPECIALITIES	9 Hours
Intensive care and Anesthesiology - safety surgery save lives - Emergency department clinical risk - Obstetric safety patient - Patient safety in internal medicine - Patient safety in Radiology.		
Module IV	REGULATORY STANDARDS FOR MEDICAL DEVICE MAINTENANCE	9 Hours
International Standards - Medical Device Directive 93/42/EEC - Medical Electrical Equipment ISO 60601- Safety Testing of Medical Devices ISO 62353 - Medical Device Inspection ISO17020 - Indian Standards - National Health Mission - Biomedical Equipment Management and Maintenance Program (BMMP), ISO 9001-2008- AERB Compliance - Radiation protection - AE(RP)R-2004- Safety Code AE/RF-MED/SC-3.		
Module V	MEDICAL ETHICS	9 Hours
Definition of Medical ethics - Scope of ethics in medicine - American medical Association code of ethics - CMA code of ethics - Fundamental Responsibilities - The Doctor and The Patient - The Doctor and The Profession - Professional Independence - The Doctor and Society - Case Studies.		
TOTAL:		45 Hours
FURTHER READING: Hospital Maintenance		
Text books:		
1.Donaldson L, Ricciardi W, Sheridan S, Tartaglia R, editors. Textbook of Patient Safety and Clinical Risk Management [Internet].		
2.Cham(CH):Springer;2021.PMID: 36315660.		
References:		
1.William Charney, Hand book of Modern Hospital Safety, CRC Press,2 nd Edition,2009.		
2.Almira Badnjevic, Mario Cifrek, Ratko Magjarevic, Zijad Dzemic, Inspection of Medical Devices: For Regulatory Purposes, Springer Nature, 2018		

2303BM016	MEDICAL WASTE MANAGEMENT					L	T	P	C			
	(B.E-BME)					3	0	0	3			
PREREQUISITE:												
	Basic Medical Sciences											
	Microbiology & Pathology Basics, Environmental Science & Chemistry											
COURSE OBJECTIVES:												
	1.Understand the classification and sources of medical waste in healthcare systems.											
	2.Learn the national and international guidelines for biomedical waste management.											
	3.Apply safe procedures for collection, segregation, treatment, and disposal of medical waste.											
	4.Evaluate the health and environmental risks associated with improper waste management.											
	5.Explore sustainable strategies for waste reduction and hospital safety.											
COURSE OUTCOMES:												
On the successful completion of the course the students will be able to,												
CO1	Classify and identify different types of biomedical waste.											
CO2	Explain biomedical waste management rules and regulatory frameworks.											
CO3	Apply waste segregation, handling, storage, and transportation protocols.											
CO4	Evaluate health and environmental risks due to improper waste disposal.											
CO5	Design sustainable and compliant waste management systems in healthcare.											
COs Vs POs MAPPING:												
COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	
CO1	3	1	-	-	-	2	2	2	-	-	1	
CO2	3	2	-	-	-	3	2	2	-	-	1	
CO3	3	3	2	2	-	3	3	2	-	-	1	
CO4	3	3	2	2	-	3	3	2	-	-	1	
CO5	3	3	3	3	2	3	3	3	-	-	1	
COs Vs PSOs MAPPING												

Biomedical Waste Management Rules, 2016 (India) and amendments, CPCB and SPCB roles and responsibilities, WHO guidelines on healthcare waste, Segregation and color-coding standards Legal obligations of generators, transporters, and operator		
Module III	WASTE HANDLING, TRANSPORTATION, AND STORAGE	9Hours
Waste collection and labelling protocols, Intra-facility and inter-facility transportation, Safe storage and time limits, Personal protective equipment (PPE), Barcoding and tracking systems, Accidental spill and exposure management		
Module IV	TREATMENT AND DISPOSAL METHODS	9Hours
Treatment methods: autoclaving, incineration, microwaving, chemical disinfection, Disposal methods: deep burial, encapsulation, shredding, On-site vs off-site treatment systems, Environmental impacts of incineration, Emission and effluent control norms		
Module V	WASTE REDUCTION AND SUSTAINABILITY IN HOSPITALS	9Hours
Waste minimization techniques: reuse, recyclable materials, green procurement, Hospital waste audit and documentation ,Occupational health and safety in waste management, Emergency preparedness and training, Integration with Sustainable Development Goals (SDGs)		
TOTAL:		45 Hours
FURTHER READING: Hospital Management		
Text books:		
1.Acharya D.B. and Meeta Singh, <i>Biomedical Waste Management in Hospitals</i> , Minerva Press, 2000.		
2.Singh, Anant Preet, <i>Biomedical Waste Management</i> , LAP Lambert Academic Publishing, 2018		
References:		
1.WHO, <i>Safe Management of Wastes from Healthcare Activities</i> , 2nd Edition, World Health Organization, 2014.		
2.CPCB Guidelines, <i>Biomedical Waste Management Rules</i> , Ministry of Environment, Forest and Climate Change, Govt. of India.		
3.Sunil Kumar et al., <i>Biomedical Waste Management: Challenges and Solutions</i> , TERI Press, 2017.		

2303BM017	PHYSIOLOGICAL MODELING							L	T	P	C
	(B.E-BME)							3	0	0	3
PREREQUISITE:											
	Biomechanics										
COURSE OBJECTIVES:											
	1.Introduce the principles of physiological system modeling.										
	2.Enable students to develop mathematical and computational models of key physiological systems.										
	3.Apply control theory and system dynamics to physiological processes.										
	4.Analyze the behavior of physiological systems under normal and pathological conditions.										
	5.Simulate physiological models using computational tools.										
COURSE OUTCOMES:											
On the successful completion of the course the students will be able to,											
CO1	Understand the fundamentals of modeling and simulation of physiological systems.										
CO2	Formulate mathematical models of physiological subsystems using first principles.										
CO3	Apply system theory and control principles to biological systems.										
CO4	Analyze the dynamic behavior of cardiovascular, respiratory, and renal systems.										
CO5	Simulate physiological models using appropriate software tools and validate with clinical data.										
COs Vs POs MAPPING:											
COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	2	-	-	-	1	-	-	-	-	1
CO2	3	3	2	-	-	1	-	-	-	-	1
CO3	3	3	3	3	2	2	-	-	-	-	1
CO4	3	3	3	3	2	2	-	-	-	-	1
CO5	2	3	3	3	3	2	-	-	-	-	1
COs Vs PSOs MAPPING											
COURSE CONTENTS:											
Module I	INTRODUCTION TO PHYSIOLOGICAL MODELING									9 Hours	
Need for modeling and systems approach, Types of models: physical, mathematical, and analog Lumped parameter and distributed parameter models, Model classification: linear vs nonlinear, deterministic vs stochastic, Steps in physiological system modeling, Compartmental models and examples											

Module II	SYSTEM PROPERTIES AND CONTROL THEORY	9 Hours
Homeostasis and feedback control in physiology, Time and frequency domain analysis System properties: linearity, time-invariance, causality Modeling using transfer functions Block diagram reduction, state-space modeling Biological control systems examples (temperature, glucose, BP)		
Module III	CARDIOVASCULAR SYSTEM MODELING	9 Hours
Heart mechanics and circulation model Wind kessel model of the arterial system Cardiac output regulation Modeling blood pressure control using baro receptor feedback Electrical analogs of blood flow Nonlinear and time-varying aspects in cardiovascular modeling		
Module IV	RESPIRATORY AND RENAL SYSTEM MODELING	9 Hours
Lung mechanics: compliance, resistance Modeling of airflow and gas exchange Control of breathing: chemical feedback model Kidney functions and fluid balance Modeling nephron and glomerular filtration Acid-base balance and osmo regulation		
Module V	SIMULATION AND VALIDATION	9 Hours
Parameter estimation techniques, Model calibration and validation using clinical data Simulation using MATLAB/Simulink or Python Sensitivity and stability analysis Case studies: patient-specific modeling, AI-assisted diagnostics Ethical considerations in physiological modeling		
TOTAL:		45 Hours
FURTHER READING: Gait analysis		
Text books:		
1.Mukkamala, R., Chon, K. H., & Khoo, M. C. K., <i>Physiological Control Systems: Analysis, Simulation, and Estimation</i> , MIT Press, 2013.		
2.Kapur, J. N., <i>Mathematical Modelling</i> , New Age International, 2015.		
References:		
1.Michael C.K. Khoo, <i>Physiological Control Systems</i> , IEEE Press, 2000.		
2.David C. Shelledy, Jay I. Peters, <i>Mechanical Ventilation</i> , Cengage Learning, 2014.		
3.R.S. Khandpur, <i>Biomedical Instrumentation: Technology and Applications</i> , McGraw-Hill		

2303BM018	PATTERN RECOGNITION TECHNIQUES					L	T	P	C		
	(B.E-BME)					3	0	0	3		
PREREQUISITE:											
	Basics of Probability and Linear Algebra										
COURSE OBJECTIVES:											
	1.To introduce the fundamentals of pattern recognition and feature extraction.										
	2.To understand various classification, clustering, and decision-making techniques.										
	3.To apply machine learning methods for pattern recognition in biomedical and real-world datasets										
	4.To evaluate and optimize models using performance metrics.										
	5.To develop hands-on skills using pattern recognition tools and applications.										
COURSE OUTCOMES:											
On the successful completion of the course the students will be able to,											
CO1	Understand the basic concepts and challenges of pattern recognition.										
CO2	Apply supervised and unsupervised classification techniques to datasets.										
CO3	Extract and select features for classification tasks.										
CO4	Evaluate and validate pattern recognition models using performance metrics.										
CO5	Implement pattern recognition algorithms for real-life biomedical or image-based problems.										
COs Vs POs MAPPING:											
COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	2	-	2	-	1	-	-	-	-	1
CO2	3	3	3	3	2	2	-	-	-	-	1
CO3	3	3	2	3	2	2	-	-	-	-	1
CO4	3	3	2	3	2	2	-	-	-	-	1
CO5	3	3	3	3	3	2	-	-	-	-	1
COs Vs PSOs MAPPING											
	CO	PSO1	PSO2	PSO3							
	CO1	2	1	-							
	CO2	3	2	-							
	CO3	3	2	1							
	CO4	3	3	2							
	CO5	3	3	2							
COURSE CONTENTS:											
Module I	INTRODUCTION TO PATTERN RECOGNITION								9 Hours		
Definitions and examples, Applications in biometrics, medical diagnosis, speech and image recognition The design cycle of a pattern recognition system Statistical decision theory Bayesian decision theory and minimum error rate classifiers											
Module II	FEATURE EXTRACTION AND SELECTION								9 Hours		
Types of features (texture, shape, color, time-series features)Linear and nonlinear transformations											

Feature normalization and standardization Dimensionality reduction: PCA, LDA Feature selection methods: filter, wrapper, embedded		
Module III	SUPERVISED LEARNING TECHNIQUES	9 Hours
k-Nearest Neighbors (k-NN) Linear classifiers, Decision trees Support Vector Machines (SVMs) Neural Networks: Perceptron, MLP basics Ensemble methods: Random Forest, Bagging, Boosting		
Module IV	UNSUPERVISED LEARNING TECHNIQUES	9 Hours
Clustering: k-Means, Hierarchical, DBSCAN Mixture models and Expectation Maximization (EM) Self-Organizing Maps (SOMs) Hidden Markov Models (HMM) – basic concepts Applications in image segmentation, gene expression clustering		
Module V	MODEL EVALUATION AND BIOMEDICAL APPLICATIONS	9 Hours
Confusion matrix, ROC curves, Precision, Recall, F1-score Cross-validation and bootstrapping Case studies: ECG pattern recognition, Tumor classification, Gait analysis, Voice recognition Real-time and embedded pattern recognition systems Introduction to tools: MATLAB, Python (scikit-learn, Tensor Flow), WEKA		
TOTAL:		45 Hours
FURTHER READING: MRI and CT Scan		
Text books:		
1. Duda, R.O., Hart, P.E., Stork, D.G., <i>Pattern Classification</i> , 2nd Ed., Wiley, 2001.		
2. Bishop, C. M., <i>Pattern Recognition and Machine Learning</i> , Springer, 2006.		
References:		
1. Sergios Theodoridis, Konstantinos Koutroumbas, <i>Pattern Recognition</i> , Academic Press, 4th Edition, 2009.		
2. Earl Gose, Richard Johnsonbaugh, Steve Jost, <i>Pattern Recognition and Image Analysis</i> , Prentice Hall, 1996		
3. Jain, A. K., Duin, R. P. W., & Mao, J., <i>Statistical Pattern Recognition: A Review</i> , IEEE Transactions, 2000.		

2303BM019		BODY AREA NETWORKS								L	T	P	C
		(B.E-BME)								3	0	0	3
PREREQUISITE:													
		Embedded Systems											
		Digital Signal Processing, and Human Physiology											
COURSE OBJECTIVES:													
		1.To introduce the fundamental concepts of Body Area Networks (BAN) and their relevance to healthcare											
		2.To explore various biomedical sensors used in BANs for vital parameter monitoring.											
		3.To understand wireless communication protocols and architectures suited for BANs.											
		4.To examine security, power, and data management challenges in BANs.											
		5.To expose students to applications and real-world implementations in smart healthcare systems											
COURSE OUTCOMES:													
On the successful completion of the course the students will be able to,													
CO1		Understand the principles, architecture, and significance of body area networks in healthcare.											
CO2		Identify suitable biomedical sensors and communication modules for BAN applications.											
CO3		Analyze the communication protocols and data transmission methods in BANs.											
CO4		Evaluate the power consumption, privacy, and security challenges in BANs.											
CO5		Design and simulate a body area network system for remote health monitoring or assistive care.											
COs Vs POs MAPPING:													
COs		PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	
CO1		3	2	-	-	-	2	-	-	-	-	1	
CO2		3	3	2	-	2	2	-	-	-	-	1	
CO3		3	3	3	2	3	2	-	-	-	-	1	
CO4		2	3	3	3	2	3	2	2	-	-	1	
CO5		3	3	3	3	3	2	2	2	-	-	1	
COs Vs PSOs MAPPING													
		COs				PSO1	PSO2	PSO3					
		CO1				2	2	-					
		CO2				3	2	1					
		CO3				3	3	2					
		CO4				2	3	2					
		CO5				3	3	3					
COURSE CONTENTS:													
Module I		INTRODUCTION TO BODY AREA NETWORKS										9 Hours	
Definition, need, and scope of BANs Components of BANs: sensors, actuators, processing unit, communication modules BAN vs PAN (Personal Area Network)Human physiological parameters that can be monitored Healthcare applications: chronic disease, elderly care, athletic performance													
Module II		BIOMEDICAL SENSORS AND SIGNAL ACQUISITION										9 Hours	

Sensors for ECG, EEG, EMG, SpO ₂ , temperature, motion, glucose Analog front-end circuit design basics Noise reduction and signal amplification Sensor placement and physiological considerations Wearable and implantable sensor design principles		
Module III	COMMUNICATION IN BANS	9 Hours
Wireless communication protocols: Bluetooth, BLE, ZigBee, UWB, Wi-Fi, NFCWBAN standards: IEEE 802.15.6 In-body, on-body, and off-body communications Network topology: star, mesh Data compression and latency optimization technique		
Module IV	POWER MANAGEMENT AND SECURITY IN BANS	9 Hours
Low-power design and energy harvesting methods Battery technologies and power-aware routing Security challenges: data encryption, authentication, privacy threats HIPAA and GDPR relevance in healthcare data transmission Fault tolerance and reliability issues		
Module V	APPLICATIONS AND CASE STUDIES	9 Hours
Smart health monitoring systems Fall detection, seizure alert systems Post-surgery rehabilitation and gait monitoring Real-time BAN simulation using NS-2/NS-3 or MATLAB Research trends in AI-integrated BANs and IoMT (Internet of Medical Things)		
TOTAL:		45 Hours
FURTHER READING: WBSN in medical application		
Text books:		
1.G. Z. Yang, <i>Body Sensor Networks</i> , Springer, 2nd Edition, 2014.		
2.Dinesh K. Sharma, <i>Wireless Body Area Networks for Healthcare Applications</i> , River Publishers, 2019.		
References:		
1.R. S. Khandpur, <i>Biomedical Instrumentation: Technology and Applications</i> , McGraw Hill, 2014.		
2.IEEE 802.15.6 Standard documentation for WBANs		
3.Axel Jantsch, <i>Design and Test of Body Sensor Networks</i> , Springer, 2020		

PROFESSIONAL ELECTIVE - IV

SL. NO.	COURSE CODE	COURSE TITLE	CATEGORY	PERIODS PER WEEK			TOTAL CONTACT PERIODS	CREDITS
				L	T	P		
1.	2303BM020	Neural Engineering	PEC	3	0	0	3	3
2.	2303BM021	Medical Device Regulations	PEC	3	0	0	3	3
3.	2303BM022	Economics and Management for biomedicalEngineers	PEC	3	0	0	3	3
4.	2303BM023	Human Assistive Devices	PEC	3	0	0	3	3
5.	2303BM024	Brain Computer Interface and Applications	PEC	3	0	0	3	3
6.	2303BM025	Virtual Reality and Augmented Reality in Healthcare	PEC	3	0	0	3	3

2303BM020	NEURAL ENGINEERING					L	T	P	C		
	(B.E-BME)					3	0	0	3		
PREREQUISITE:											
	Human anatomy										
COURSE OBJECTIVES:											
	1.To understand the basics of nervous systems and its functions										
	2.To characterize neuronal cells										
	3.To identify the effect of neuronal functions and the neural disorders										
	4.To select suitable testing methods for analyzing neural behaviour										
COURSE OUTCOMES:											
On the successful completion of the course the students will be able to,											
CO1	Apply the concept of action potential propagation to understand neuron Synaptic potentials and Receptor potentials										
CO2	Analyse the characteristics of neuronal cells based on its characterisation										
CO3	Analyse the effect of neuronal function based on the neuro transmission models										
CO4	Analyse the different types of neurological disorders and its causes										
CO5	Analyse the neuronal behavioural using appropriate testing methods										
COs Vs POs MAPPING:											
COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	1	2	3								1
CO2	2	3	2								1
CO3	2	3	2								1
CO4	2	3	2								1
CO5	2	3	2								1
COs Vs PSOs MAPPING											
				COs	PSO1	PSO2	PSO3				
				CO1	1	1	1				
				CO2	1	1	1				
				CO3	2	1	1				
				CO4	2	1	1				
				CO5	2	1	2				
COURSE CONTENTS:											
Module I	INTRODUCTION TO NERVOUS SYSTEM									9 Hours	
Nervous system: Introduction, Central and peripheral nervous system, Signalling molecules, First growth factor, First Neuro transmitters in brain, functional organization, Synaptic potentials and Receptor potentials.											
Module II	NEUROANATOMY									9 Hours	

Structures and functions of neurons, Synapse: function, signals produced by neurons, Sensors function, Glialcells, molecular and cellular organization of neuronal differentiation, characterization of neuronal cells.		
Module III	NEURO PHYSIOLOGY AND NEURO PHARMACOLOGY	9 Hours
Resting and action potentials; Mechanism of action potential conduction; Voltage dependent channels; nodes of Ranvier; Chemical and electrical synaptic transmission. Synaptic transmission, neurotransmitters and their release; fast and slow neurotransmission; characteristics of neurites; hormones and their effect on neuronal function		
Module IV	NEUROLOGICAL DISORDERS	9 Hours
Pathogenesis, Genetic basis of neurological disorders, Psychiatric Disorders: Psychiatric epidemiology, Unipolar depression, Bipolar depression, Seasonal affective disorder, Panic disorder, Autism, Stroke, Huntington disease		
Module V	NEURONAL MECHANISM	9 Hours
Neuronal mechanism of behaviour, Animal behaviour, Behaviour in various environments, Behavioural and cognitive neuroscience, Behavioural studies using animal model, Testing motor functions, Grip Strength Test, Testing Cognitive Functions, Learning and memory related test		
TOTAL:		45 Hours
FURTHER READING: Clinical Neurology		
References:		
1.Georg Goldenberg, Bruce L.Miller-Neuro psychology and Behavioral Neurology_ Handbook of Clinical Neurology, Elsevier - libgen.lc., 2008		
2.MichaelJ.Aminoff, Handbook of Clinical Neurology, Elsevier, London, 2012		
3.MasonP.,Medical Neurobiology, Oxford University Press, 2011		
4.Mathews G.G.Neurobiology,2 nd edition, Black well Science, UK, 2000		
5.Gordon M. Shepherd G.M, and Shepherd Neurobiology, 3rd Edition Oxford University Press, USA, 1994		

2303BM021		MEDICAL DEVICE REGULATIONS						L	T	P	C	
		(B.E-BME)						3	0	0	3	
PREREQUISITE:												
		Biomedical Engineering, Human Anatomy & Physiology										
		Medical Device Design principles										
COURSE OBJECTIVES:												
		1.To introduce the regulations in medical device design										
		2.To discuss the regulations of medical device design in various medical industries										
COURSE OUTCOMES:												
On the successful completion of the course the students will be able to,												
CO1		Understand the basic concepts of medical device regulations										
CO2		Apply the global policies on medical device regulations										
CO3		Analyze implications of the regulations										
CO4		Analyze the Standards and Regulations used for medical devices										
CO5		Analyze the software and Quality system regulation in medical device design										
COs Vs POs MAPPING:												
COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	
CO1	-	2	-	2	-	1	1	2	-	-	1	
CO2	-	2	-	2	-	1	1	2	-	-	1	
CO3	-	3	-	2	-	1	1	2	-	-	1	
CO4	-	3	-	2	-	1	1	2	-	-	1	
CO5	-	-	-	2	-	1	1	2	-	-	1	
COs Vs PSOs MAPPING												
				COs	PSO1	PSO2	PSO3					
				CO1	1	2	-					
				CO2	1	2	-					
				CO3	1	2	-					
				CO4	1	2	-					
				CO5	1	2	-					
COURSE CONTENTS:												
Module I		INTRODUCTION									9 Hours	
Defining the device, Overview of quality function deployment, Business proposal Reliability: Definition, Quality Vs Reliability Vs Unreliability, Types of Reliability, Optimizing reliability, Reliability’s effects on medical devices. Concept of Failure: Causes of Failure, Practical aspects of failure, Failure rates, Hardware failure, Software Failure. Safety and Risk Management: Medical device safety and risk management, Effectiveness/performance of medical devices, Phases in the life span of a medical device												
Module II		DRUG MANUFACTURING PRACTICES									9 Hours	
Global Harmonization Task Force (GHTF): Objectives, Scope of the four GHTF study groups, Benefits of the GHTF, Global Medical Device Nomenclature (GMDN) The Food and Drug Administration: Device classification, Registration and listing, The 510 (k) Process, Declaration of conformity, The PMA application, Investigational Device Exemptions (IDEs), Good Manufacturing Practices(GMPs).												

Module III	MEDICAL DEVICE DIRECTIVES	9 Hours
The European Union: European Directives, European Standardization Bodies, European Standards Development Process, Other European Standards Considerations, Conformity Assessment and Testing, European Organization for Testing and Certification. The Medical Devices Directives: Process, Choosing the appropriate directive, Identifying the applicable essential requirements		
Module IV	STANDARDS AND REGULATIONS	9 Hours
Standards and Regulation: Voluntary and mandatory standards, Standards development process, Conformity assessment with standards, National and international standards systems, Identification of standards, Current trends in the use of standards in medical device regulations. The ISO 9000 Series of Standards		
Module V	SOFTWARES AND QUALITY SYSTEM REGULATIONS	9 Hours
Software and Quality system regulation: Software as a Technology, Domestic and International Software Regulations and Standards. Design controls, Document controls, Purchasing controls, Identification and traceability, Production and process controls, Acceptance activities, Non-conforming product, Corrective and preventive action		
		TOTAL:
		45 Hours
FURTHER READINGS: Medical ethics		
Text books:		
1. Michael Cheng, Medical Device Regulations Global Overview and Guiding Principles, World Health Organization, 2003.		
2.Des O'Brien, Medical Device Regulations Roadmap A Beginners Guide, Create Space Independent Publishing Platform, 2017		
References:		
1. Aakash Deep, Medical Device Regulations A Complete Guide, Elsevier Science, 2022.		
2. Jack Wong, Raymond Tong, Jenny Stanford Publishing Handbook of Medical Device Regulatory Affairs in Asia, Second Edition, 2018.		
3. G.R Higson, Medical Device Safety, The Regulation of Medical Devices for Public Health and Safety, 2001.		

2303BM022	ECONOMICS AND MANAGEMENT FOR BIOMEDICAL ENGINEERS					L	T	P	C		
	(B.E-BME)					3	0	0	3		
PREREQUISITE:											
	Basic knowledge of healthcare systems and engineering project processes										
COURSE OBJECTIVES:											
	1.To introduce fundamental concepts of economics relevant to healthcare and biomedical industries										
	2.To understand cost analysis and budgeting in biomedical product development										
	3.To apply principles of management in healthcare and biomedical projects										
	4.To explore marketing and entrepreneurship opportunities in biomedical engineering										
	5.To promote strategic decision-making, resource allocation, and project planning for biomedical professionals										
COURSE OUTCOMES:											
On the successful completion of the course the students will be able to,											
CO1	Understand basic economic concepts and their application in biomedical industries										
CO2	Analyze the cost, pricing, and budgeting strategies in healthcare technology development										
CO3	Apply project management and operations strategies in biomedical engineering										
CO4	Evaluate healthcare systems from an economic and managerial perspective										
CO5	Demonstrate entrepreneurial thinking and marketing strategies in biomedical innovation										
COs Vs POs MAPPING:											
COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	-	-	2	-	2	2	2	-	-	1
CO2	3	2	3	3	3	2	2	2	-	-	1
CO3	3	2	3	3	3	2	3	2	-	-	1
CO4	3	2	2	3	2	3	3	3	-	-	1
CO5	2	2	3	3	3	3	2	3	2	2	1
COs Vs PSOs MAPPING											
	COs	PSO1	PSO2	PSO3							
	CO1	2	2	-							
	CO2	3	3	-							
	CO3	3	3	2							
	CO4	3	3	2							
	CO5	2	3	3							
COURSE CONTENTS:											
Module I	INTRODUCTION TO HEALTH ECONOMICS								9 Hours		
Nature and scope of health economics Supply and demand in healthcare Economic evaluation: Cost-benefit, cost-effectiveness, cost-utility analysis Health insurance models Public vs private healthcare financing											
Module II	BIOMEDICAL PRODUCT COSTING AND BUDGETING								9 Hours		

Cost elements in medical device manufacturing Activity-based costing for biomedical projects Break-even analysis Budget planning and variance analysis Return on investment (ROI) in healthcare innovation		
Module III	PRINCIPLES OF MANAGEMENT IN HEALTHCARE	9 Hours
Functions of management: Planning, Organizing, Leading, Controlling Decision making and leadership in biomedical organizations Hospital operations and resource management Supply chain management in biomedical logistic Quality management systems (QMS) and Lean Six Sigma in healthcare		
Module IV	MARKETING AND ENTREPRENEURSHIP	9 Hours
Biomedical product lifecycle Marketing strategies for medical devices and servicesMarket research, segmentation, branding Entrepreneurial ecosystem in healthcare Regulatory aspects for biomedical startups		
Module V	STRATEGIC AND PROJECT MANAGEMENT	9 Hours
Strategic planning in healthcare organization Project lifecycle: Initiation to closure Tools: Gantt charts, CPM, PERT Risk assessment and mitigation in biomedical projects Case studies on large-scale health-tech project execution		
TOTAL:		45 Hours
FURTHER READING: Hospital management		
Text books:		
1.Michael H. Merson, Robert E. Black, Anne J. Mills, <i>Global Health: Diseases, Programs, Systems, and Policies</i> , Jones & Bartlett Learning.		
2.Charles E. Beggs, <i>Engineering Management</i> , Routledge, 2017.		
3..M. Pandey, <i>Financial Management</i> , Vikas Publishing		
References:		
1.Peter F. Drucker, <i>Management: Tasks, Responsibilities, Practices</i> , HarperBusiness.		
2.Dinesh Mohan & O.P. Goel, <i>Economics of Healthcare Systems</i> , PHI Learning.		
3.N. M. Wadia, <i>Entrepreneurship and Management in Healthcare</i> , Tata McGraw-Hill		

2303BM023	HUMAN ASSIST DEVICES (For B.E-BME)								L 3	T 0	P 0	C 3
PREREQUISITE:												
Biomedical instrument												
COURSE OBJECTIVES:												
1.Comprehend the assistive technology(AT) used for mobility 2.Summarize the AT for sensory impairment of vision and hearing 3.Uncover the assist devices for vital organs and advancements in AT 4.Improving the hearing and respiratory comprehension 5.Maintain the functionality of trending devices for disabled												
COURSE OUTCOMES:												
CO1	Develop innovative and indigenous technologies to enhance the quality of life and inclusion of persons with disabilities.											
CO2	Demonstrating the appropriate use and strengths and limitations of disability-specific technologies											
CO3	Design, application, and advancements of assistive devices for vital organs, integrating engineering, medical technology, and ethical considerations.											
CO4	Analyze the design of respiratory systems and hearing aids.											
CO5	Design of new devices used by trending technologies.											
COs Vs POs MAPPING:												
COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	
CO1	3	2	3	2	3	2	3	2	-	-	1	
CO2	3	2	2	2	2	3	3	2	-	-	1	
CO3	3	3	3	3	3	2	3	2	-	-	1	
CO4	3	2	3	3	2	2	2	2	-	-	1	
CO5	3	3	3	3	3	2	2	3	-	-	1	
COs Vs PSOs MAPPING												
				COs	PSO1	PSO2	PSO3					
				CO1	3	2	1					
				CO2	3	2	1					
				CO3	3	3	2					
				CO4	3	3	2					
				CO5	3	3	3					
Module I		ASSISTIVE TECHNOLOGY FOR MOBILITY									9 Hours	
Basic assessment and evaluation for mobility – Control systems, navigation in virtual space by wheel chairs- Wheel chair seating and pressure ulcers- Fuzzy logic expert system for automatic tuning of myoelectric Prostheses- Intelligent prosthesis												
Module II		ASSISTIVE TECHNOLOGY AND SENSORY IMPAIRMENTS									9 Hours	
Visual and auditory impairment, assessment methods - Libraille, GRAB, mathematical Braille – Augmentative and alternative methods for hearing impairment – Use of multimedia technology to help hard of hearing children – Haptic as a substitute for vision.												
Module III		ASSIST DEVICES FOR VITAL ORGANS AND ADVANCEMENTS IN TECHNOLOGY									9 Hours	
Cardiac assist devices, Intra-Aortic Balloon Pump(IABP)- auxiliary ventricles – Dialysis for kidneys, Intermittent positive pressure breathing - (IPPB) type assistance for lungs – Latest use of assistive technology for chronic heart diseases and healthcare – Information technology, telecommunications, new media in assisting healthcare – Future trends in assistive technology – virtual reality-based training system for disabled children												

Module IV	RESPIRATORY AND HEARING AIDS	9 Hours
Ventilator and its types- Modes of ventilation- Breathing apparatus operating sequence- Electronic IPPB unit with monitoring for all respiratory- Deafness- Hearing aids- Wearable devices for hearing correction.		
Module V	TRENDS OF HUMAN ASSIST DEVICES	9 Hours
Transcutaneous electrical nerve stimulator (TENS), TENS used to treat, TENS techniques - Bio Feedback – Diagnostic and point of care platforms (POC)- Prevailing status of POC tools in the Indian healthcare system- The Utility of POC diagnostic devices in India.		
Total:		45Hours
FURTHER READINGS: Medical devices design		
Text Books:		
1.Brownsell, Simon,etal. “ <i>A systematic review of lifestyle monitoring technologies,</i> ”Journal of telemedicine and telecare 17.4 (2011): 185-189		
2.Marion.A.Hersh,MichaelA.Johnson,“ <i>AssistiveTechnology for visually impaired and blind</i> ”		
3.Authors: Mr. K. Gopal Ram, Mrs. D.K. Kalaivani, Mr. G. Pratheep and Dr. M. Sivaprakash		
ISBN: 978-81-968148-7-8 DOI: https://doi.org/10.59646/humanasdev/097 Date of Publication: January 06, 2024		
References:		
1.Daniel Goldstein, MehmetOz,“ <i>CardiacassistDevices</i> ”,Wiley,2000.		
2.The book of “ Human Assist Devices” ISBN 13 9786207998555 Publication Date - 28-08-2024		
Publisher - KS Omni Scriptum Publishing		

2303BM024	BRAIN COMPUTER INTERFACE AND APPLICATIONS	L	T	P	C						
	(B.E-BME)	3	0	0	3						
PREREQUISITE:											
	Basic knowledge of human neurophysiology and signal processing										
COURSE OBJECTIVES:											
	1.To understand the fundamentals of brain signal acquisition and processing.										
	2.To explore brain-computer interface (BCI) system architecture and paradigms.										
	3.To study algorithms used in neural signal decoding and translation.										
	4.To analyze various clinical and non-clinical BCI applications.										
	5.To evaluate ethical and design aspects of BCI systems in rehabilitation and assistive technology										
COURSE OUTCOMES:											
On the successful completion of the course the students will be able to,											
CO1	Explain the basic structure and functioning of BCI systems and neural mechanisms.										
CO2	Acquire and preprocess EEG signals for BCI applications.										
CO3	Apply signal processing and machine learning techniques for BCI signal classification.										
CO4	Analyze and design real-world BCI systems for clinical and assistive applications.										
CO5	Evaluate ethical concerns, limitations, and future directions in BCI technologies.										
COs Vs POs MAPPING:											
COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	2	2	2	1	-	2	2	-	-	1
CO2	3	2	3	3	3	-	2	2	-	-	1
CO3	3	3	3	3	3	-	2	2	-	-	1
CO4	3	3	3	3	3	2	2	3	2	2	1
CO5	3	2	2	2	2	2	3	3	-	2	1
COs Vs PSOs MAPPING											
	COs	PSO1	PSO2	PSO3							
	CO1	3	2	-							
	CO2	3	3	2							
	CO3	3	3	3							
	CO4	3	3	3							
	CO5	2	2	3							
COURSE CONTENTS:											
Module I	INTRODUCTION TO BRAIN-COMPUTER INTERFACE				9 Hours						
Introduction and definition of BCI Neuro anatomy: Brain regions and functions Types of BCIs: Invasive, semi-invasive, non-invasive BCI system components and architecture Neural signals: EEG, ECoG, MEG, Fnirs Neuro plasticity and BCI training											
Module II	BRAIN SIGNAL ACOUISITION AND PREPROCESSING				9 Hours						

EEG acquisition techniques: Electrodes, placement, impedance Signal characteristics and artifacts Noise removal: Filtering, ICA, PCA Event-Related Potentials (ERP), P300, SSVEP, Mu rhythms Real-time data acquisition tools and protocols		
Module III	SIGNAL PROCESSING AND FEATURE EXTRACTION	9 Hours
Time-domain, frequency-domain, and time-frequency methods Feature extraction techniques: Power spectral density, wavelet transform Feature selection and dimensionality reduction Machine learning classifiers: SVM, LDA, k-NN, deep learning basics Real-time classification and feedback mechanisms		
Module IV	BCI APPLICATIONS AND CASE STUDIES	9 Hours
Medical applications: Motor rehabilitation, communication in ALS, stroke, locked-in syndrome Neuro feedback and mental state monitoring Assistive technologies: Prosthetic control, wheelchair navigation Gaming and virtual reality-based BCIs Brain-to-brain communication concepts		
Module V	CHALLENGES, TRENDS AND ETHICS	9 Hours
BCI reliability, speed, accuracy, and robustness Ethical issues: Privacy, consent, cognitive liberty BCI in cognitive enhancement and military uses Hybrid BCIs and multimodal interface Recent advancements and open research challenges		
TOTAL:		45 Hours
FURTHER READING: Human anatomy		
Text books:		
1.Wolpaw J. R., Wolpaw E. W., <i>Brain–Computer Interfaces: Principles and Practice</i> , Oxford University Press, 2012.		
2.Rajesh P. N. Rao, <i>Brain–Computer Interfacing: An Introduction</i> , Cambridge University Press, 2013		
References:		
1.Jonathan Wolpaw & Dennis J. McFarlane, <i>The Future of BCI: Research and Applications</i> , Elsevier, 2020.		
2.Ramesh Srinivasan, <i>EEG Signal Processing for BCI</i> , CRC Press, 2021.		
3.Farina, Dario & Gazzoni, Marco, <i>Bioelectrical Signal Processing for BCI and Neuroprosthetics</i> , Academic Press, 2021		

2303BM025	VIRTUAL REALITY AND AUGMENTED REALITY IN HEALTHCARE					L	T	P	C		
	(B.E-BME)					3	0	0	3		
PREREQUISITE: Human Anatomy & Physiology											
Medical Terminology & Clinical Procedures, Signal Processing Basics											
Human-Computer Interaction (HCI)											
COURSE OBJECTIVES:											
1.To introduce the relevance of this course to the existing technology through demonstrations, case studies and applications with a futuristic vision along with socio, economic impact and issues											
2.To understand virtual reality, augmented reality and using them to build Biomedical engineering applications.											
3.To know the intricacies of these platform to develop PDA applications with better optimality											
COURSE OUTCOMES:											
On the successful completion of the course the students will be able to,											
CO1	Assess the principle and functions of Virtual Reality(VR)										
CO2	Apply modeling procedures to develop VR applications										
CO3	Design and configure Haptic mode line regard to human parameters										
CO4	Assess the principle and components of Augmented Reality(AR)										
CO5	Analyse utility of computer vision and techniques in Augmented Reality(AR)										
COs Vs POs MAPPING:											
COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	2	3		2							1
CO2	2	2	3	2							1
CO3	2	2	3	2							1
CO4	2	2	3	2							1
CO5	2	3	2	2							1
COs Vs PSOs MAPPING											
COs	PSO1	PSO2	PSO3								
CO1	2	2									
CO2	2	2									
CO3	2	2									
CO4	2	2									
CO5	2	2									
COURSE CONTENTS:											
Module I	INTRODUCTION TO VIRTUAL REALITY								9 Hours		
Definition of Virtual Reality (VR),Principles of VR, Main components, the five classic components of a											

VR system - Input Devices: (Trackers, Navigation, and Gesture Interfaces): Three-dimensional position trackers, navigation and manipulation-interfaces and gesture Interfaces-Output Devices: Graphics displays-sound displays & haptic feedback, Physiological data recording, Position and movement measuring systems, Problems in VR.		
Module II	VR DEVELOPMENT PROCESS & CONTENT CREATION CONSIDERATIONS	9 Hours
Geometric modelling - kinematics modelling- physical modelling - behaviour modelling - model Management. Methodology and terminology-user performance studies-VR health and safety issues-Usability of virtual reality system- cyber sickness -side effects of exposures to virtual reality environment		
Module III	VR ON THE WEB&VR ON THE MOBILE	9 Hours
JS-pros and cons-building blocks (Web VR, Web GL, Three.js, device orientation events)- frameworks (A-frame, React VR)- Google VR for Android-Scripts, mobile device configuration, building to android-cameras and interaction-teleporting-spatial audio-Assessing human parameters-device development and drivers-Design Haptics.		
Module IV	INTRODUCTION TO AUGMENTED REALITY WITH AR HARDWARE	9 Hours
Defining augmented reality, history of augmented reality, Augmented Reality Hardware – Displays – Audio Displays, Haptic Displays, Visual Displays, Other sensory displays, Visual Perception, Requirements and Characteristics, Spatial Display Model. Processors – Role of Processors, Processor System Architecture, Processor Specifications. Tracking & Sensors - Tracking, Calibration, and Registration, Characteristics of Tracking Technology, Stationary Tracking Systems, Mobile Sensors, Optical Tracking, Sensor Fusion		
Module V	COMPUTER VISION FOR AR& ARTECHNIQUES	9 Hours
Computer Vision for Augmented Reality - Marker Tracking, Multiple-Camera Infrared Tracking, Natural Feature Tracking by Detection, Simultaneous Localization and Mapping, Outdoor Tracking Augmented Reality Software - Introduction, Major Software Components for Augmented Reality Systems, Software used to Create Content for the Augmented Reality Application, AR Techniques – Marker based and Marker Less approach.		
TOTAL:		45 Hours
FURTHER READING: Augmented Reality, Mixed Reality, and Hybrid Approach in Healthcare Simulation		
Text book:		
1.Jason Jerald.2015.TheVR Book :Human, Centred Design for Virtual Reality. Association for Computing Machinery and Morgan & Claypool, New York, NY, USA. 2.C. Burdea & Philippe Coiffet, Virtual Reality Technology, Second Edition, Gregory, John Wiley & Sons, Inc.,2008.		
References:		
1.Augmented Reality: Principles and Practice (Usability) by Dieter Schmalstieg & Tobias Hollerer, Pearson Education (US), Addison, Wesley Educational Publishers Inc, New Jersey, United States, 2016. ISBN: 9780321883575. 2.Wadee Alhalabi, Virtual Reality Implementation in Healthcare Settings, Medical Information Science Reference, 2017. 3.James Roland,Virtual Reality and Medicine, Reference Point Press, Incorporated,2018.		

2303BM025		ADVANCEMENTS IN HEALTHCARE TECHNOLOGY								L	T	P	C
		(B.E-BME)								3	0	0	3
PREREQUISITE:													
		Human Anatomy & Physiology, Biomedical Instrumentation, Medical Imaging, and Healthcare Information Systems											
COURSE OBJECTIVES:													
		1.Understand the needs for wearable devices and the technology											
		2.Learn the concepts in digital health care and digital hospitals											
		3.Apply the tools in design, testing and developing digital health care equipment											
COURSE OUTCOMES:													
On the successful completion of the course the students will be able to,													
CO1	Interpret the need for digital methods of handling medical records												
CO2	Explain the digital radiology												
CO3	Modify the tools and methods for work flow in E-Health												
CO4	Identify the available technology for wearable healthcare devices												
CO5	Compare various standards for inter-operability of devices, quality and safety standards for developing healthcare systems												
COs Vs POs MAPPING:													
COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11		
CO1	3	2	2	2	-	-	-	-	-	-	1		
CO2	3	2	2	2	-	-	-	-	-	-	1		
CO3	3	2	2	2	-	-	-	-	-	-	1		
CO4	3	3	2	2	-	-	-	-	-	-	1		
CO5	3	3	2	2	-	-	-	-	-	-	1		
COs Vs PSOs MAPPING													
				COs	PSO1	PSO2	PSO3						
				CO1	-	2	-						
				CO2	-	2	-						
				CO3	-	2	-						
				CO4	-	2	-						
				CO5	-	2	-						
COURSE CONTENTS:													
Module I		DIGITAL HEALTH										9 Hours	
Digital Health: Requirements and best practices, Laws and regulations in Digital health, Ethical issues, barriers and strategies for innovation.													
Module II		DIGITAL RADIOLOGY										9 Hours	
Digital radiology for digital hospital, picture archiving and communication, system integration, digital history of radiology, medical image archives, storage and networks.													
Module III		E-HEALTH										9 Hours	

E-Health: Health care networking, medical reporting using speech recognition, physiological tests and functional diagnosis with digital methods, tele-consultation in medicine and radiology.		
Module IV	M-HEALTH CARE AND WEARABLE DEVICES	9 Hours
Introduction to mobile healthcare devices-economy-average length of stay in hospital, outpatient care, health care costs, mobile phones, 4G, smart devices, wearable devices, Uptake of e-health and m-health technologies. Standards, system Design and case study.		
Module V	MODALITY AND STANDARDS FOR INTER-OPERABILITY	9 Hours
Multimodality registration in daily clinical practice. Mobile healthcare. Selection and Implementation in e-Health project, design of medical equipment based on user needs. Security and privacy in digital health care. Case study.		
TOTAL:		45 Hours
FURTHER READINGS: Wearables and The Internet of Things for Health		
Text books:		
1.Christoph Thuemmler, Chunxue Bai, “Health 4.0: How Virtualization and Big Data are Revolutionizing Healthcare”, Springer, 1st ed. 2017		
2.Wlater Hruby, “Digital revolution in radiology - Bridging the future of health care, second edition, Springer, New York. 2006		
3.Samuel A. Fricker, Christoph Thummler , Anastasius Gavras, “Requirements Engineering For Digital Health”, Springer, 2015		
References:		
1.Rick Krohn (Editor), David Metcalf, Patricia Salber, “Health-e Everything: Wearables and The Internet of Things for Health, ebook. 2013.		
2.Khandpur,R.S,”Handbook of Biomedical Instrumentation ”,Second Edition. Tata Mc Graw Hill Pub. Co., Ltd. 2003		
3.John, G. Webster. Medical Instrumentation: Application and Design. Second Edition. Wiley Publisher, New Delhi. 2013.		

2303BM038		HOSPITAL MANAGEMENT								L	T	P	C
		(B.E-BME)								3	0	0	3
PREREQUISITE:													
COURSE OBJECTIVES:													
		1. To understand the fundamentals of hospital administration and management.											
		2. To explore various information management systems and relative supportive services.											
		3. To learn the quality and safety aspects in hospital.											
COURSE OUTCOMES:													
On the successful completion of the course the students will be able to,													
CO1		Explain the principles of Hospital administration.											
CO2		Identify the importance of Human resource management.											
CO3		List various marketing research techniques.											
CO4		Identify Information management systems and its uses.											
CO5		Understand safety procedures followed in hospitals.											
COs Vs POs MAPPING:													
	COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	
	CO1	3	2	1	-	-	1	1	-	1	-	1	
	CO2	3	2	1	-	-	1	1	-	1	-	1	
	CO3	3	2	1	-	-	1	1	-	1	-	1	
	CO4	3	2	1	-	-	1	1	-	1	-	1	
	CO5	3	2	1	-	-	1	1	-	1	-	1	
COs Vs PSOs MAPPING													
							COs	PSO1	PSO2	PSO3			
							CO1	1	1	-			
							CO2	1	1	-			
							CO3	1	1	-			
							CO4	1	1	-			
							CO5	1	1	-			
COURSE CONTENTS:													
Module I		HOSPITAL MANAGEMENT									9 Hours		
Nature and Scope of a Hospital, History of Indian Hospitals, Distinction between Hospital and Industry, Challenges in Hospital Administration, Hospital Planning- Equipment Planning- Functional Planning - Current Issues in Hospital Management – Telemedicine - Bio-Medical Waste Management.													
Module 1I		HUMAN RESOURCE MANAGEMENT IN HOSPITAL									9 Hours		
Human Resource Management – Principle, Characteristics, Functions, Significance and Importance - Profile of HRD Manager, Good HR Practices, Causes for Poor Human Resource Management, Tools of HRD, Human Resource Inventory – Manpower Planning, Recruitment, Selection, Induction, Training Guidelines, Promotion, Termination and Communication.													
Module III		CHALLENGES IN HOSPITAL MANAGEMENT									9 Hours		
Managing A Service Organization - Hospital Service Delivery – Quality Control – Six Sigma, NABH Hospital Queuing Systems – Simple Queuing Systems, Interdependent Queuing Systems, - Hospital. Management Functions – Operation Management, Finance and Cost Management, HR Management, Materials Management - Case Studies													
Module IV		SERVICES IN HOSPITAL									9 Hours		
- patient care (Emergency, Surgery, Pediatrics, Cardiology, Oncology), diagnostics (Radiology, Lab/Pathology, Blood Bank), support (Pharmacy, IT, HR, Finance, Housekeeping, Dietary), and specialized areas (ICU, Maternity, Psychiatry, Physiotherapy)													
Module V		HOSPITAL INFORMATION SYSTEMS & SUPPORTIVE SERVICES									9 Hours		

Management Decisions and Related Information Requirement - Clinical Information Systems - Support Service Technical Information Systems – Medical Transcription, Medical Records Department – Central Sterilization and Supply Department – Pharmacy– Food Services - Laundry Services.		
	TOTAL:	45 HOURS
Text books:		
1. Ramani K V, “Hospital Management-Text and Cases”, Pearson education, New Delhi,2013. 2. Malhotra A K, “Hospital Management - An Evaluation, Global Publications”, New Delhi, 2009.		
Learning Materials:		
1. R.C.Goyal, “Hospital Administration and Human Resource Management”, Seventh Edition, PHI, 2017. 2. G.D.Kunders, “Hospitals – Facilities Planning and Management”, TMH, New Delhi, Fifth Reprint 2017.		